Procurement Technical Specification For Denver RTD EMU Communication System

H-Rotem
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1. Introduction

1.1 General

This Procurement Technical Specification (PTS) defines the technical requirements for Communication system to be purchased by Hyundai Rotem Company and installed in the Electric Multiple Unit (EMU) Commuter Rail Cars, identified as the "Fastrack Eagle Project" cars, to be manufactured by Hyundai Rotem and to be operated by the DENVER Regional Transportation District("Owner"). The cars will be operated by DENVER RTD over the DENVER Metropolitan Area Railroad(East Corridor, Gold Line, Northwest Rail).

1.2 General Requirements

It is the intent that all Suppliers providing equipment or services to Hyundai Rotem shall comply with the Customer's requirements. When the Customer does not have any specific requirements, the Supplier shall comply with the requirements of this PTS as appropriate. Suppliers must comply with the requirements stated herein unless otherwise agreed to in writing by Hyundai Rotem. Any conflict between the Customer's requirements and this specification shall be brought to Hyundai Rotem's immediate attention for resolution. Notwithstanding conflict resolution, the most restrictive requirement shall apply. The term "Customer", as used in this document, refers to Hyundai Rotem's Customer (DENVER RTD).

This PTS in no way relieves the Supplier from any requirements specified in the Technical Specification.

All formal communication between Hyundai Rotem staff and Suppliers must be through the Procurement function at Hyundai Rotem. Formal transmittal from Suppliers of drawings, procedures, reports, and analyses referenced in this document must be sent to the responsible Hyundai Rotem Procurement Representative defined in the instruction letter transmitting this PTS. Formal approvals and documentation from Hyundai Rotem to Suppliers will be sent via the responsible Procurement Representative.

The order of precedence for Supplier technical requirements shall be as follows:

1) The Purchase Order between Hyundai Rotem and the Supplier

2) The Contract between Hyundai Rotem and the Customer

3) Hyundai Rotem's PTS

4) DENVER RTD Technical Specification Contract

As a guide, the sections of the aforementioned documents are considered to contain specific and relevant requirements have been referenced in this document. However, this in no way diminishes the Suppliers' responsibility for compliance with the Technical Specification. The complete requirements are those found within this PTS document, the Technical Specification, the Contract, addendums, and the documents referenced therein. It shall be the Supplier's responsibility to assure that current revisions of the subject documents are being utilized, and to assure that equipment, documentation, and services furnished pursuant to this PTS are in full compliance with all applicable documents.

1.3 Train Configuration

The configuration of on unit is as follows:

(a) Total number of train; 25 train sets.

(b) Each train(Unit) will consists of MA and MB.

2. Definitions and Abbreviations

This section includes a list of all definitions and abbreviations applicable to PTS, GS and TS.

2.1 Definitions

The following definitions are applicable to the PTS, Technical Specification for the Purchase of DENVER RTD (TS).

"Contractor" means the persons or person appointed by the Employer to undertake the execution of the works for Hyundai-Rotem. In order to avoid misunderstanding the roles of the Contractor in GS and TS, the term "Contractor" shall be read as "Subcontractor" for those clauses referred to in the GS and TS by the PTS.

"Engineer" means the personnel appointed by DENVER RTD for the purpose of the DENVER RTD Electrical Multipule Unit.

"Subcontractor" means a supplier of complete Communication System to Hyundai-Rotem and Contractor specified in TS with regard to Communication system. Subcontractor shall carry out the works in accordance with TS with regard to Communication system.

2.2 Abbreviations

The following abbreviations shall apply to the PTS and TS.

GS: General Specification for the purchase of Electrical Multiple Unit Commuter Railcars.

TS: Technical Specification for the purchase of Electrical Multiple Unit Commuter Railcars.

PTS: Procurement Technical Specification for DENVER project issued by Hyundai-Rotem

RTD: Regional Transportation District

NA: Not applicable

EMI/EMC : Electro Magnetic Interference / Compatibility

S.A : System Assurance

RAMS : Reliability Availability Maintainability System safety

PDR : Pre-Design Review

FDR : Formal Design Process

3. Scope of supply and Work

3.1 Hardware

The Subcontractor shall be responsible for the provision of all equipments for the Communication System. The system shall comply the DENVER RTD specification (CONTRACT PART I. 3.17.1) and the following specification or more. Subcontractor shall provide all components related to Communication system, but not limited to, the followings:

Comment ;
우진의 공급 범위는 아니며 I/F 만 포함.

(1) Radio

1-1) Integrated radio control unit.

1-2) ~~Mobile~~ VHF radio transceiver.

1-3) Rooftop antenna.

1-4) Interconnecting wiring shall be Teflon high temperature jacketed type RG-303/U MIL-C-

17D coaxial cable with approved connectors.

1-5) Radio power supply unit with suitable interference suppression line filter. 우진의 공급 범위가 아니다.

1-6) Suitable apparatus in the DENVER control center for the train dispatcher to receive and display this car number information. 우진의 공급 범위는 아니며 I/F 협의 대응.

~~1-7) Radio-transmit control pushbutton, located on the right side of the cab center console outboard of brake controller. If possible, should be redundant at left hand.~~

1-8) Programmable Automatic-Time-Out (ATO) function

1-9) Audio recorder (Section12.1(a) of Attachment 7, Part C)

2) Public Address (PA) shall measure the local ambient noise level between use and automatically compensate gain accordingly and instantly just prior to every announcement.

2-1) Push-to talk button

2-2) Speaker selection Switch (3 positions)

2-3) Microphone

2-4) Power Amplifier

2-5) Speakers

3) Automatic Vehicle Location System (AVL)

3-1) Global Positioning Satellite System (GPS)

3-2) Wireless LAN (WLAN)

3-3) Dead Reckoning System for operation within the 1.5mile center City Commuter Connector tunnel plus other short highway overpass sections, which will provide the geographical location of the vehicle.

3-4) Vehicle Location Control Panel in each cab.

4) Passenger Emergency Intercom (PEI), one at each of the two wheelchair locations and one at the doorway area at the end of the car opposite the wheelchair locations.

Each cab shall be equipped with two passenger emergency intercom system momentary contact pushbuttons or rocker switches, identified as "Passenger Intercom Push to Talk" and "Passenger Intercom Reset".

5) Communicating Buzzer System

6-1) One at each Conductor's doorway control panel, both upper and lower panels, for a total of 8.

6-2) One in each cab, for a total of either 1 or 2 per car.

6-3) One located at each end of the car, adjacent to the body end, for a total of 2.

6-4) One located on the car exterior at each low level doorway, easily reached when standing on a low level station platform, for a total of 4.

6-5) One located on the car exterior on each side of the end of the car, at each cab end of a car, easily reached when standing on the car during coupling/uncoupling, for a total of either 2 or 4 per car.

6) Cab Text Messages

7) Wireless Network

Comment ;

8) Wayside Installations 우진의 공급 범위는 아니며 I/F 협의 대응.

9) Communication Control Panel in each Cab, which has full function (control of PA, IC PEI and buzzer, and all indicator LEDs to signal activation)

10) Conductor Communication Control Panel at Each doorway, which has PA, IC and buzzer function. This shall have separate momentary contact switches and indicator LEDs for control of the PA and IC functions. This panel shall also include a low profile, noise canceling microphone, and an IC speaker, both behind a perforated grille. A communication buzzer and push button shall also be included.

11) Key switch

12) Front Destination Indicator

13-1) Front, one exterior end destination

13-2) Side destination indicator shall be provided at each side doorway, for a total of 4 per car.

13-3) Interior, at least 6 display units. Two is at the front end of each section.

(* Note: The interior destination sign system shall be configured to permit the future connection to an active message system to be provided by DENVER for current and news and other information. And the subcontractor shall provide the interface schematic diagram for the active message system for carbody wiring).

13-4) Destination Sign Control Panel can be combined with Vehicle Location Control Panel

13) Accessory; All accessory for the communication system shall be provided by the subcontractor. The accessory includes pug connectors, pins or sockets, special cable for communication system, and fixing bolts & nuts for wayside equipment, and etc.

H/R Comments (24/11/2010)

1) Design shall be based on the altitude conditions and climatic conditions of the Denver area.
2) Equipment to be housed in equipment houses shall be rated for an operating temperature range of -40°F ~ 160°F at 0% to 95% relative humidity, non-condensing.

Comment ; -40°F 저온 동작 성능 보증은 IC/LCD 등 소자 특성에 따라 별도 협의한

3.2 Software

The Subcontractor shall provide, as a minimum, the followings:

1) Automatic Announcement

The Subcontractor shall provide all source codes, application software, mapping software and hardware required to make initial and future announcement changes, and shall disclose to all required proprietary methods, software, tools and associated equipment necessary to ensure the successful operation of the system. All required licenses required to permit DENVER RTD to operate, maintain and modify the system for the lifetime of the car shall be included at no additional cost.

2) Programming Equipment for Radio

Five complete sets of radio programming equipment shall be supplied for setup and maintenance of the car radio transceiver by DENVER RTD. The subcontractor shall comply the detail requirement in the TS.

3) Destination Sign System

The subcontractor shall provide all source codes, software and hardware required to make initial and future sign system revisions, and shall disclose all required proprietary methods, software, tools and associated equipment necessary to ensure the successful operation of the system. All required licenses or other fees required to permit DENVER RTD and Hyundai-Rotem to operate, maintain and modify the system for the lifetime of the car shall be included at no additional cost to Hyundai-Rotem. All details of the destination signs system and operation of the control system shall be presented to Hyundai-Rotem and the Engineer of DENVER RTD.

4) Automatic Vehicle Location System

The subcontractor shall prepare all mapping and survey work (location database) necessary to establish trigger points for all announcements, and prepare all software for complete operation. All mapping and survey work performed pursuant to this contract shall be the property of Hyundai-Rotem and DENVER RTD, and shall be delivered to Hyundai-Rotem in a format that is transportable and/or convertible in furtherance of DENVER RTD's interests. The subcontractor shall supply all required software and special equipment by a site license agreement as part of this Contract. The subcontractor shall provide all source codes, mapping software and hardware which is required to make initial and future announcement changes, shall prepare and install all updates and revisions to the cars for the basic warranty period for each car, and shall disclose all required proprietary methods, software, tools and associated equipment necessary to ensure the successful operation of the system. All required licenses required to permit to operate, maintain and modify the system for the lifetime of the car shall be included at no additional cost. There shall be no additional or reoccurring costs incurred by Hyundai-Rotem for any software, documentation, or special equipment required to operate, program, update, change displays and/or announcements and maintain the system as delivered. This shall include all software and hardware necessary to incorporate future changes and revisions in routes, destination displays, announcements, public relations messages and so forth.

The system in all operating modes shall indicate vehicle location to within 3 meters horizontally at 95% probability and the WAAS-enabled GPS system shall provide accuracy sufficient for track identification. The system and components shall be designed to permit integration with current digital and analog-type radio systems and cell phone networks. The subcontractor shall provide all source codes, mapping software and hardware along with a complete bill of material. The subcontractor shall disclose to DENVER RTD all required proprietary methods, software and tools and associated equipment necessary to ensure the successful operation of the system. All aspects of the system shall be in compliance with the Americans with Disabilities Act as amended.

5) Expansion Provisions

An Automatic Vehicle Location (AVL) and announcement system utilizing an open architecture shall be supplied to provide for future expansions and upgrades to the system. The subcontractor shall provide all source codes; mapping software and hardware required to make initial and future changes, and shall disclose all required proprietary methods, software, tools and associated equipment necessary to ensure the successful operation of the system.

6) Wireless Local Area Network

All WLAN systems, the Denver RTD Control Center and selected management department shall be connected via a secure extranet. Associated servers and networking software shall be supplied by the subcontractor. Extranet fees be borne by the subcontractor for the full term of the contract (last car of warranty).

7) Non-commercially available software

The subcontractor shall provide the all non-commercially available software defined in the TS.

8) Commercially Available software

All commercially available software shall be listed-up by the subcontractor.

3.3 Wayside Installation

The Subcontractor shall provide all hardware and engineering work for the wayside equipment, as a minimum, the following: Comment :

(1) Radio 우진의 공급 범위는 아니며 T/F 협의 대응

The subcontractor shall provide, install, test and verify all of the equipment in the DENVER RTD Control Center required to receive, decode and display the data transmitted by the vehicle radio

system required under DENVER RTD technical specification. The subcontractor shall install in the Control Center all equipment required to allow the ten train dispatcher consoles to communicate to train passengers via the public address system on the cars. This will allow the train dispatcher to give public service announcements directly to the passengers. This shall be done by selecting a particular car or cars, all cars or all cars on a selected route, using apparatus approved by the Engineer. There shall be no more than a 2 second delay for any variation of car selection prior to the system permitting the announcement to be made. Details shall be submitted to the Engineer for approval.

2) CAB Text Messages

The subcontractor shall design and supply a complete system to permit electronic transmission of written Form movement permits from the DENVER RTD EMU dispatcher to the Operator in the cab during all times when the train is in operation. This shall include the selection of proper radio transmission methods, the installation of all equipment in the DENVER RTD Control Center and all other wayside apparatus as required, all carborne equipment, and all licenses, permits and other registration fees required for a period of at least 20 years from the delivery of the last car.

Comment : 우진의 공급 범위는 아니며 지구 협의 대응.

3) Wireless Local Area Network

As part of this Contract of Denver RTD project, the subcontractor also supply and install wireless local area networks (WLAN) in DENVER RTD project. (including all hardware, software, extranet and monitoring location equipment as follows):

Individual WLAN locations shall be installed at each of DENVER RTD's Railroad Division five (5) yards and shops along with one (1) location logistically central to train operations such as DENVER RTD's Suburban Station, for the primary purpose of automatic fault data exchange between DENVER RTD center and the DENVER RTD EMU along with automatic file update downloads for the AVL system. The facility based WLANs shall also accommodate other maintenance management and information software used by DENVER RTD so they are accessible via the WLAN networks.

All WLAN location data systems shall be configured to automatically capture the car number, fault and status information from each Denver RTD EMU as it enters within any WLAN area. All fleet status information shall be integrated into a real-time information software system with monitoring capabilities at each yard, the DENVER RTD Control Center and one key management location through a common extranet provided by the Contractor.

The subcontractor shall design the topography of the network as applicable for each location and supply and install all needed hardware and software. The IEEE802.11 technology chosen for the WLAN shall be based on the site surveys for network coverage concerning reception such as signal-to-noise ratios, EMI interferences, speed potential and estimated user bandwidth loads. An adequate number of wireless access points and associated antenna shall be provided to accommodate efficient speed and access based on the use of the proposal automated data transfer system, DENVER RTD's Vehicle Maintenance Information System (VMIS) and Vehicle Technical Information Library (VTIL) system.

3.4 Spares, Special Tools and Testing Equipment

The Contractor shall furnish the following capital spares, test equipment, and special tools. If a Rolling Stock Option is exercised, the proportional number of capital spares will be provided. Selected special tools and test equipment shall also be required depending on the nature of the tools/equipment and its intended usage.

3.4.1 Special Tools and Testing Equipment

All tools, excluding tools commonly available from standard suppliers such as Granger and/or Snap-On that are required or desirable for the effective/efficient conduct of maintenance and

repair of the vehicles will be provided as a "special tool" in quantities necessary to support the level and intensity of vehicle maintenance anticipated. This requirement extends through mechanical, electrical, electromechanical, and electronic tools, including software products. These tools will be provided within 60 days of the delivery of the first production Car of the Base Order.

All test equipment required or desirable for the conduct of maintenance and repair activities of the Car will be provided in sufficient quantity to adequately support the maintenance operation. All such test equipment to be provided by the Contractor regardless of the equipment's availability from other common sources. Portable test equipment (Two of each kind, including instruction manuals, cables, connectors and associated software), shall be provided for all on-board systems. All portable test equipment shall be delivered within 30 days of the delivery of the first production Car of the Base Order. One set of bench test equipment shall be provided for the purpose of testing, troubleshooting and calibrating all electrical, electronic, mechanical and electro-mechanical components of each vehicle subsystem. All test equipment, including bench test equipment, shall be delivered within 90 days of the delivery of the first production Car of the Base Order.

The Subcontractor shall provide, at a minimum, the following:

The spare parts shall be based on the principal that local maintenance will be to Level One. This level assumes the diagnosis and replacement of major parts and line replaceable units (LRU) locally. Defective units and/or their sub-assemblies will be returned to the subcontractor for repair.

The Subcontractor shall recommend and submit the detailed list of spare parts for maintenance.

The Subcontractor may recommend additional spare parts and separately quote spare parts if he esteems necessary for maintenance and repair of the supplied Lighting for Hyundai Rotem's further consideration as separate purchase.

The Subcontractor shall propose the spare parts according to Section 20.12 Spare Parts and provide the following parts as a minimum, according to proposed in BAFO submission.

3.4.2 Spare

A minimum of 2 Car sets of all replaceable/rebuildable interior body units will be provided as capital spares deliverable to the Site before the delivery of the last production Car of the Base Order. This provision will include but not be limited to all interior body panels and fixtures, seats, window assemblies, windshields assemblies door leafs, end doors, and other body parts as may be damaged and require replacement.

A minimum of 3 Car sets of all replaceable/rebuildable mechanical/electrical units will be provided as capital spares deliverable to the Site before the delivery of the last production Car of the Base Order. This provision will include but not be limited to fully dressed (built-up, ready to install) trucks, traction motors, air compressors, air conditioning compressors, air conditioning condensers, complete air conditioning units (including frame), propulsion equipment, auxiliary power equipment, train control equipment, communications/radio equipment, safety equipment, master controller, operators "desk" equipment, door operators and other major mechanical, electrical, and electromechanical equipment.

A minimum of 4 Car sets of couplers and draw bar assemblies will be provided as capital spares deliverable to the Site before the delivery of the last production Car of the Base Order. In addition, 4 compromise couplers will be provided.

H/R Comments (24/11/2010)

1) SUB CONTRACTOR must supply H/R all the accessories including but not limited to spares, special tools, test equipments and etc.

3.5 Interface Responsibilities

The subcontractor shall provide recommendations and specifications on the installation of all the subcontractor's supply equipment. The subcontractor shall provide the interface specification between Communication system and vehicle equipments (as TS, but not limited to) and performed the interface test between Communication system and vehicle equipments. Hyundai Rotem will perform the installing design and installing work of Communication system with the subcontractor's information. The subcontractor shall develop and provide a performance criteria which shall include such items as maximum allowable percentage of false reports and maximum allowable percentage of missed failure. The subcontractor shall provide the weight and the power consumption of Communication systems.

. H/R Comments (24/11/2010)

1)SUB CONTRACTOR must provide Monthly Report that contains following :
- Work that was performed by SUB CONTRACTOR for the month covered by the report
- Work that will be performed by SUB CONTRACTOR for the coming month

3.6 Design Information

The Subcontractor shall provide all necessary documents and deliverables for Hyundai-Rotem according to the time schedule defined by Hyundai-Rotem. The Subcontractor shall provide the interface information, which possibly affects performance, fitting, and form, from Hyundai-Rotem. The documents and the time schedule are listed and specified in part II of this specification "Design Information".

The Subcontractor shall provide the technical requirement and design information. (refer to section 4 and 5 of this PTS)

All drawing shall be provided by Auto-cad file.

3.7 Testing and Inspection

The Subcontractor shall be responsible for performance tests of the Communication System. The subcontractor shall be responsibility for a compressive series to tests to be performed to verify the suitability of design and workmanship of each car according to the requirement in DENVER RTD TS requirement (CONTRACT PART I, SECTION 4). And the subcontractor shall provide the test procedure and the test report after perform the each test. Hyundai Rotem and/or DENVER RTD have the right to witness any of these tests at the any stage of test progress.

Hyundai Rotem and/or DENVER RTD will carry out the First Article Inspection (FAI) on complete Communication System under the Subcontractor's responsibility. The Subcontractor shall carry out the type test and routine test of equipment and assembly. The commissioning test of the complete vehicle will be performed by Hyundai Rotem under the Subcontractor's responsibility for the Communication SYSTEM and coupling assembly. The Subcontractor shall provide the required information for testing and carry out the tests and retrofit problems during the tests in accordance with PTS.

In the event that any test for the Communication SYSTEM is failed, the Subcontractor shall, at his own expense, take whatever action is deemed such as, rectification, readjustment or design changes to the satisfaction of Rotem and DENVER RTD, in order to meet the testing requirements. The individual equipments, systems and sub-systems, shall be type- and routine-tested in accordance with detailed respective test procedures to be drawn up by Contractor and agreed by the HR which shall take into account the requirements of respective standard required on the Denver specification and-/-or- other appropriate international standards, and tests program drawn up by the Contractor to demonstrate that the individual equipments, sub-systems and systems meet the other specified requirements and agreed by the HR and Denver Engineer. Such tests may be performed either at the Contractor's works, or on site, as appropriate, and as agreed with the Engineer.

All such tests shall be carried out at the Contractor's cost, wherever performed, in the presence of and to the satisfaction of the HR and Denver Engineer, who reserves the right to witness any or all of

the tests including FAI.

Wherever any equipment, system or sub-system is not specifically covered by an internationally recognized specification or test procedure, or where the type and routine tests prescribed by North American regulations or IEC or other international standard do not adequately cover the requirement, tests which are acceptable both to the Contractor and to the HR/Denver Engineer, shall be devised.

Type test and FAI for individual equipment may be waived if these were carried out earlier on equipments of identical design, witnessed by a reputed organization, and the service performance of such equipments was found to be reliable. The Contractor shall submit a proposal in this regard to the Denver Engineer for review. The waiver of Type Test is entirely at the discretion of the Denver Engineer.

Without prejudice to any other provisions of the Contract, the HR/Denver Engineer reserves the right to witness any or all of the tests, and to require submission of any or all test specifications and reports. The their Engineer reserves the right to reasonably call for additional tests as are considered necessary, including the quality of welds particularly in highly stressed areas, by non destructive testing methods. Prototype tests may be required to verify the suitability of the process or the materials proposed. Their Engineer may if considered necessary may call for conducting optional tests as per relevant standards without any additional cost to the contractor.

The all results of all tests (type and routine) shall be submitted to the HR's Engineer, who will record his conclusions as to whether or not the equipment being tested has passed satisfactorily.

H/R Comments (20/12/2010)

1) Contractor shall be responsible for design and verification/commissioning in accordance with HR's provide material workmanship guideline attached.

The contactor shall provide compliance statement against HR's material workmanship guide line during design stage.

Material_Workmanship.pdf

3.8 Operation & Maintenance Manual

The Subcontractor shall provide necessary requirements for Operation and Maintenance Manual and Spare Parts Catalogues for the Communication system and special tool with hard copies and electronic format.

The requirement for Operating and Maintenance Manuals and Spare Parts Catalogues shall be provided for Approval by Hyundai Rotem according to the time schedule approved by Hyundai Rotem. The Subcontractor shall provide softcopy for all O&M Manual according to detail requirement (Attachment 10) and required formats provided from Hyundai Rotem.

H/R Comments (24/11/2010)

1) SUB CONTRACTOR shall provide the operation and maintenance manuals in accordance with TP023 provided, DENVER RTD DOCUMENTATION AND TRAINING REQUIREMENTS TABLE and DENVER RTD DOCUMENTATION AND TRAINING REQUIREMENTS.

Denver RTD Attachment 10 .pd

3.9 Training

The Subcontractor shall provide comprehensive training to the DENVER RTD Employer's staff (maintenance, operating, training instructor and engineering) in accordance with DENVER RTD TS. The training activities and works for the communication system shall be approved by Hyundai Rotem. The subcontractor shall prepare the training manual and provide it. The training manual shall follow the sample in the Appendix A "SAMPLES" of TS

H/R Comments (24/11/2010)

1) SUB CONTRACTOR shall provide training (document & information) in accordance with TP23 DENVER RTD DOCUMENTATION AND TRAINING REQUIREMENTS TABLE and DENVER RTD DOCUMENTATION AND TRAINING REQUIREMENTS.

3.10 Warranty

The subcontractor has sole responsibility under this Contract to maintain sufficient warranty spare parts in this stock to support the warranty period requirements, according to the DENVER RTD CONTRACT PART I, SECTION 5 or supply of equipment manufactured specifically for DENVER RTD.

4. Technical Specification

The Subcontractor, who will supply the communication system for Denver project, shall comply all sections related with Communication system of Technical Specification for the purchase of Electrical Multiple Unit Commuter Railcars (TS).

4.1 Technical Requirements

The communication system can meet all technical requirement required in DENVER Technical Specification (TS).

The subcontractor shall review the below section of TS, as a minimum, and comply;

4.2 Electromagnetic Compatibility

The subcontractor shall ensure that the subsystem and components operate correctly and reliably within the Line operating electromagnetic environment, and shall not themselves materially and adversely affect adjacent third party infrastructure and systems. The subsystem shall be electromagnetically compatible in the intended railway environment and with adjacent installations.

The Subcontractor shall comply with all relevant standards and legislation, including the following standard.

- IEC 62236-3-2 Railway applications - Electromagnetic compatibility √

4.2.1 EMC Control

The Contractor shall submit electromagnetic compatibility control and test plan for approval and the plan shall insure that proper emphasis will be placed on the control of interference and adequate attention given to interface and Federal Communications Commission (FCC).

The subsystem shall operate without either suffering or causing interferences which may impact the system operation and/or safety because of conducted and/or radiated emissions and the contractor

shall employ design techniques, construction methods, and whatever equipment is required to prevent interference caused by internal sources from affecting the proper operation of the car subsystem. In addition to coordinating frequencies, the Contractor shall provide necessary on-board balancing, filtering, shielding, modulating techniques and isolation to maintain signal to noise (S/N) ratios above limits. Shielding, isolating, balancing and grounding shall be used as required to reduce the undesirable effect of interference. Electrostatic and magnetic shielding methods shall be employed to minimize the effect of stray signals and transient voltages on interconnecting cables. Interconnecting power and signal cables shall be physically separated.

Cellular (wireless) telephones, typical farecards, credit/debit cards, laptop computers, PDAs and other magnetic storage media such as floppy disks shall not be adversely affected by EMI fields at any location inside and outside the vehicle, including the floor, ceiling, Operator's cab, or in proximity of equipment cabinets. For farecards, an exposure limit of 500 Gauss shall be considered as the standard for farecard susceptibility.

The Contractor shall ensure that any static or alternating fields generated by the subsystem shall not interfere with the correct operation of cardiac pacemakers, customer goods or any magnetic media. At a minimum the magnetic flux density in any car due to the subsystem shall not exceed the ICNIRP guideline elsewhere under any mode of operation, including failure modes.

4.2.2 Emission Limitations

The Contractor shall ensure that the cars do not produce any radiated emissions that will violate FCC Rules and Regulations, 47 CFR Part 15.103. In addition, for purposes of design validation, the subsystem shall not produce radiated emission levels above those permitted by Figure 10-2(UM05), Curve 2 of MIL-STD-461E when measured 100 feet from the track centerline. The radiated emissions of the Pilot Car shall not exceed the level permitted by the above standard at any frequency between 0.15 MHz and 1000MHz. Testing shall be based on the document "Radiated Interference in Rapid Transit Systems, Volume 2 - Suggested Test Procedures, UMTA-MA-06-0153-85-11" for guidance and shall be conducted to the satisfaction of the Engineer.

4.2.3 Safety Analysis

The Contractor shall prepare an EMI safety analysis which demonstrates the protection against hazards due to EMI. A draft analysis shall be submitted for review at the early Preliminary Design Review; an updated version shall be submitted at the mid-point in the design process between the Preliminary Design Review and the Final Design Review and the completed analysis shall be submitted at the early Final Design Review.

The Contractor shall document through the EMI safety analysis that the subsystem's EMI emissions are safe under normal conditions, and that the subsystem provides adequate detection, annunciation, and response to failures which could cause EMI to increase above acceptable levels. The EMI safety analysis shall include a Preliminary Hazard Analysis; a Fault Tree; and a Failure Mode Effects

Analysis.

The EMI safety analysis shall document the protective actions which prevent a hazard occurrence. The EMI safety analysis shall demonstrate that the equipment and its actions are adequate to prevent an EMI hazard. The EMI safety analysis shall distinguish between failures which affect EMI and for which there is automatic protection, those which are annunciated for Operator or Maintainer action, and those which are undetected failures.

The EMI safety analysis shall consider all relevant failures of the subsystem, including failures of line filter components, EMI detection and annunciation systems, and any other components that can affect the level or frequency of the generated EMI; unusual operating conditions; and failures of wayside equipment affecting, such as a substation component failure. The EMI safety analysis must also include equipment self-testing in areas that affect EMI.

4.2.4 Compatibility Design Report

The Contractor shall prepare an Electromagnetic Compatibility (EMC) Design Report prior to Final Design Review, and subsequently update it when and if design changes occur as a result of testing. The report shall consist of two parts, Emissions Report and Susceptibility Report, as described below.

The Emissions Report shall document:

a) EMC emission control methods, including equipment layout, circuit routing, frequency coordination and stability, filters, interconnections, grounding, and shielding. The Contractor shall comply with IEEE Standard 1100-1992, Recommended Practice for Powering and Grounding Sensitive Electronic Equipment, unless otherwise approved;

b) Description of each unit with a peak power rating of greater than 5 kW, including the emission characteristics versus operating mode, voltage and loading, as well as line filter, output filter, line input impedance, and circuit and operating mode considerations;

c) Characteristics of any repeating narrow band emissions and frequency stability control measures.

The Susceptibility Report shall document:

a) EMC susceptibility control methods, including equipment layout, equipment enclosure design, circuit routing, frequency coordination and stability, filters, interconnections, grounding, and shielding;

b) The high voltage and low voltage grounding arrangement, and provisions for worst-case ground or return bus voltages within the car;

d) The normal and fault paths for ground currents;

e) The equipment protection as well as any restrictions on equipment operation or maintenance necessary to ensure susceptibility control, such as not leaving covers off of equipment enclosures; and

f) For each type of electrical interface between subsystems or enclosures:

i) The driving circuit, including impedance and maximum and minimum output voltage and current levels;

ii) The receiver circuit, including maximum and minimum voltage or current thresholds, margin against error, and destructive threshold;

iii) All aspects of the connection requirements, such as single wire, double wire, twisted pair, shield, and shield grounding; and

iv) Any special circuit or connection provisions needed to ensure EMC.

Comment ;
화재 규격을 만족하는 제품 사용. Technical Provision Section 20.25에
만족하여 재질을 선택/적용한다.

4.3 Fire

4.3.1 Fire Safety

The fire safety analysis for each equipment shall be submitted in accordance with all provisions of 49 CFR 238.103 (c), Fire Safety Analysis for Procuring New Passenger Equipment, and APTA RP-PS-005-00, Fire Safety Analysis of Existing Passenger Rail Equipment.

4.3.2 Flammability, Smoke Emission and Toxicity Requirement

The sub-supplier shall submit a list of combustible materials intended to be used in the Car. The sub-supplier shall test combustible materials used in Cars to demonstrate compliance with the requirements set forth in 49 CFR 238.103 and NFPA 130.

The sub-supplier shall submit fire test reports from an independent laboratory indicating successful testing and demonstrating compliance with such requirements. Testing shall be conducted after the Phase 1 Effective Date and on a production batch of material intended to be used on the Car. Each laboratory shall have tested a standard test sample no greater than 30 days prior to performing the tests, the results of which shall be submitted to Hyundai-Rotem.

Materials used in Car construction, except for materials used in small parts such as knobs, rollers, fasteners, clips, grommets, and small electrical parts that would not contribute significantly to fire propagation or to smoke or toxic gas generation and are distributed throughout the Car, shall be tested for toxicity using Boeing Specification Support Standard BSS-7239. Materials shall comply with the following maximum toxic gas release limits (ppm):

Carbon Monoxide (CO)	3,500 ppm
Hydrogen Fluoride (HF)	200 ppm
Nitrogen Dioxide (NO2)	100 ppm
Hydrogen Chloride (HCL)	500 ppm
Hydrogen Cyanide (HCN)	150 ppm
Sulfur Dioxide (SO2)	100 ppm

Tests shall be conducted in flaming mode after 240 seconds using a National Bureau of Standards smoke density chamber for sample combustion. Gas sampling may be conducted during the smoke density test.

All fire test results for the material used to DENVER project must be submitted. The previous fire test of other project results cannot be allowable.

4.3.3 Electrical Fire Safety

Electrical equipment, wiring and apparatus shall conform to the standards of NFPA 130, Section 8.

4.3.4 Fire Load Calculation

Fire load calculation for all combustible material must be calculated with heat value tested in accordance with ASTM E 1354.

4.3.5 Fire Performance Deliverables

The Fire Performance Deliverables shall be provided with the following documents:

No	Submission Document	Remark

No	Submission Document	Remark
1	Fire Safety Analysis for each equipment	Reference standard - 49 CFR 238.103 (c) and APTA RP-PS-005-00
2	Fire Test Report for flammability, Smoke Emission and Toxicity	The summary table for FST test shall be delivered with test report. Refer to the attached combustible material matrix
3	Combustible material matrix	Refer to the attached combustible material matrix
4	Heat Release Rate test report	Report tested in accordance with ASTM E 1354
5	Description of design in specific areas of fire risk	

*Attachment Combustible Material Matrix

Combustible Material Matrix

4.4 RAMS

4.4.1 Reliability Requirement

The Subcontractor shall meet the specified reliability targets with annual design mileage of 125,000 miles per Car

Equipment	MDBSF (car mile)	MDBF (car mile)
Communication System	3,571,429	Not Applicable

Table 1: Reliability Targets

For failure definition, please refer to Section 5 in End User Specification Part I Scope of Work.

4.4.2 Fleet Defects (Pattern Failures)

The occurrence of independent failures of the same warranted item that exceeds more than 10 percent of the total number of identical items supplied under the subcontract may be declared a fleet defect or pattern failure. On this basis, the subcontractor shall be required to develop and implement a corrective action program to eliminate the pattern failure. The corrective action program must be approved by Rotem/End User and its implementation must be coordinated with Rotem.

4.4.3 Maintainability Requirement

1) Subcontract shall propose and demonstrate MTTR (Mean Time To Repair) based on the previous project through maintainability analysis.
2) The equipment shall be designed and constructed to have a minimum period of 92 days between periodic maintenance servicing by shop personnel.
3) The Schedule Preventive Maintenance Plan will include a designation of inspection intervals conformant with all FRA requirements, and include the inspection and maintenance standards, service intervals, wear limits, settings and tolerances.
4) Each maintenance task shall be defined in the 'Scheduled Preventive Maintenance Plan' in the format provided by Hyundai-Rotem. The 'Scheduled Preventive Maintenance Plan' shall be in compliance with the maintenance requirements specified in the Maintenance Manuals prepared by the supplier. The 'Scheduled Preventive Maintenance Plan' shall provide all information necessary for identification and quantification of the required maintenance resources.(Refer to Appendix IV)
5) Each maintenance task shall be followed by reference numbers on the maintenance manual for Hyunai-Rotem's review.
6) At hte supplier's cost, in RTP's Depot, in coordination with Hyundai-Rotem, the supplier shall demonstrate the maintainability. (Part1_5.3.Maintainability). The procedures used in the demonstration shall be the same as those included in the manuals delivered.
7) With the exception of consumables (motor brushes, brake shoes etc), overhaul period of each equipment shall be decided considering operating hours, cycles and running distance.
8) Items without a specified maintenance or overhaul period shall be designed not to require replacement during the vehicle design life.

4.4.4 Regulation and Standard Requirement

Subcontractors shall comply with 49 CFR Part 238.105, APTA's manual, MIL-STD-882C, MIL-STD-1629 and CENELEC standard EN50128 specified in attached file.

RAMS approach.pdf

4.4.5 Safety Requirement

1) The Subcontractor shall ensure that the Risk Index of all hazards on the train shall not be worse than the "Acceptable" category i.e. either "Ar" or Ac" as defined in Appendix II of attached RAMS guideline.
2) The subcontractor shall comply with the general safety design requirement specified in Section 4.3 of Mil-STD-882C.
3) Hazards of Category I or II severity shall be demonstrated by test, the validity of any portion of all analyses.

4.4.6 Failure Analysis

In the event of a failure of any component during contractually required testing or the warranty period, the subcontractor shall submit a complete failure analysis report to Rotem for approval. The failure analysis shall be submitted within 30 days of receipt of the failed part by Subcontractor, unless otherwise agreed upon. Unless otherwise specified, the format of failure report shall be mutually agreed.

4.4.7 Deliverables

1) Subcontractor shall submit Hardware and Software Safety Program and its supporting analysis to comply with 49CFR 238.105 for all micro-based safety related systems by **within 2 weeks** after order by Rotem and it shall include the contents specified in Section 4.3 in 49CRF 238.105

2) Subcontractor have to submit the Safety Integrity Level Justification Document in order to justify the appropriate SIL rating and for approval.

3) The System Hierarchy including LRU (Line Replaceable Unit) and LLRU (Lowest Line Replaceable Unit) list using the attached table by **within 2 weeks** after order by Rotem.(Refer to Appendix I)

4) The subcontractor shall prepare and submit Reliability and Maintainability Analysis Report at Preliminary Design Review and Final Design Review stage and the contents of it shall consist of, but not be limited to, the following elements

   - Definition of Reliability and Maintainability goals and criteria
   - Subsystem functional description including subsystem schematics and functional flow diagrams
   - A summary of reliability and maintainability prediction results and remedial actions taken as a result of the Reliability and Maintainability Analysis
   - A comparison of reliability and maintainability predictions against allocated requirements specified in Section 1.1 and 1.2
   - The format for reliability analysis, Maintainability analysis shall be provided in Appendix III, IV and V.

5) The subcontractor shall prepare and submit Failure Modes and Effects and Criticality Analysis at Preliminary Design Review and Final Design Review stage

6) Fault Tree Analysis including source data shall be submitted at Preliminary Design Review and Final Design Review stage.

The Hazards of Category I or II severity shall be subject to Fault Tree Analysis.

7) The subcontractor shall prepare and submit Safety Analysis Report at Preliminary Design Review and Final Design Review stage and the contents of it shall consist of, but not be limited to, the following elements.

- Definition of safety goals and criteria
- Subsystem functional description including subsystem schematics and functional flow diagrams
- Description of Safety Feature
- A summary of the objectives, findings and remedial actions taken as a result of the Safety Analysis
- Hazard Analysis including Preliminary Hazard Analysis (PHA), System Hazard Analysis if identified, Sub-system Hazard Analysis, Operating and Support Hazard Analysis using attached format(Appendix VII, VIII, IX and X)

8) The subcontractor shall provide information regarding Safety and Security Certification and Threat and Vulnerability Analysis and resolution (Refer to End User Specification_ Attachment 9 Project and Construction Management)

4.5 Software

4.5.1 SEPTA Requirement

Subcontractor shall comply with the following SEPTA's requirement.

- Clause XXXV and XXXVI of SEPTA's RFP
- Clause 1.6.5, 14.11.1, 16.14, 18.8, 18.13, 18.14, 20.8.7, 20.19.2, 20.19.3 of SEPTA Technical Specification
- SEPTA Hardware and Software Safety Program(Drawing 19-18): Subcontractor shall submit Hardware and Software Safety Program

Subcontractor shall submit following software documents as CDRL submissions.

- CDRL 16-19 Microprocessor Systems & Software Documents into Escrow
- CDRL 16-20 Alternative Microprocessor Bus Structure
- CDRL 16-21 Non-Commercial Software Documentation
- CDRL 16-22 Software Escrow Account
- CDRL 16-23 Software Quality Assurance Plan
- CDRL 16-24 Changes to Software Submittals
- CDRL 16-25 Commercially Available Software List
- CDRL 16-27 All software
- CDRL 16-28 All Software Development and Modification Tools
- Software Documentation of 16.14 of SEPTA Technical Specification
- Complete car equipment software, on CD-Rom
- CDRL 20-44 Master Software List

For single documentation methodologies, Subcontractor shall follow the contents of each software documents guided in international standard as follows.

Title	Contents	Remark
Software Quality Assurance Plan	Clause 4 of IEEE-STD-730	
System Functional Description		

Software Requirement Specification	Clause 5 of IEEE-STD-830	
Software Design Description	Annex A of IEEE-STD-1016 ATA No.102	
Software Verification and Validation Plan	Clause 7 of IEEE-STD-1012	
Software Verification and Validation Report	Clause 7.6 of IEEE-STD-1012	
Software Configuration Management Plan	Clause 4 of IEEE-STD-828	
Software User Documentation	Clause 5 of Software User Documentation	

All software supplied for the system designed structurally and logically, fully documented, thoroughly tested in a systematic manner such that they can achieve high quality in terms of safety, reliability, testability, traceability and maintainability. Also all software supplied for the system shall be developed in line with latest ISO 90003

Subcontractor have to submit the SIL Justification Document in order to justify the appropriate SIL rating and for approval.

For the adoption of EN50128 with SIL 1 and above, ISA shall perform software assessment in compliance with clause 14 of EN50128. Finally, Subcontractor shall submit final Software Assessment Report so that Rotem can ensure that assigned SIL has been achieved.

Subcontractor has to comply with Rotem's Software Quality Assurance Plan which will be approved by Customer

Especially, clause 16.14.7 (Escrow Account), the Escrow Agreement should be made among depositor (Subcontractor), Beneficiary (Customer) and Escrow Agent. Subcontractor shall have all responsibility of making arrangement for Escrow Agreement including all fees for all Escrow activities(but customer pay for escrow agent fees), discussion with Customer and etc. Subcontractor must comply with Customer's proposed standard escrow agreement document(e.g. issued by IronMountain). The deposit escrow period is 30 years after completion of warranty period. For more detailed requirement of escrow, please refer to section 2.3 of this SEMS requirement.

Subcontractor shall provide the properties and all requested material for Rotem's software quality & safety audit to Subcontractor. Corrective action against NCR and OBS should be returned to Rotem within one week after software audit. And also, Implementation of corrective actions against to Identified NCR (Non conformance Report) and OBS (Observations) should be finalized within one-month after software audit. Rotem will not pay the amount of money for software development without the completion of the identified NCR and OBS.

The Subcontractor shall be obliged to take care of any software change if Rotem and Customer's request the change for the correction of software during commissioning phase, commissioning and Warranty Period, etc. The change shall be implemented by the Subcontractor. In the event, when any non-conformity arises to the specified requirements, the Subcontractor shall take remedial measures at its own cost within the schedule to be agreed with Rotem.

The Subcontractor shall be obliged to abide by the software change control procedure on site(or online based change control system (http://sccp.rotem.co.kr) which should be prepared by Rotem in order to enhance the regulation of software change in train-set. It should be not allowed to upload the changed software into train-set without the approval by Rotem/Customer under software change control by Rotem. Subcontractor has to supply test report of system integration testing or bench testing of the updated software performed subcontractor's premises before formal release of the updated software.

During Test & commissioning stage, subcontract must inform the software version number and it's

revision history document on the monthly basis.

The Subcontractor shall provide Diagnostic & Maintenance Software and its user manual.

4.5.2 Denver Requirement

The following requirement shall be applied in addition to SEPTA's requirements.

A. Subcontractor must comply with the section 42 of Contract Part III(General Terms-Standard) for Intellectual Property Right.

B. Comply with section 3.21.3 of Contract Part I(Scope of Work) and 16.3 of Attachment 7 to the Concession Agreement.

C. The following CDRL specified in Attachment 6 of Concession Agreement shall be submitted

- CA-093(Source Code Escrow Agreement), 7B-16(Software Management Control Plan), 9-08(Software Requirement Manual)

D. Comply with 9.3.3 of Attachment 7 of Concession Agreement

E. Comply with 2.5(c) of Attachment 9 of Concession Agreement

4.5.3 Details of Escrow Requirement

Regarding the requirement of supply of source code and work product specified in section 41.2.4 and 42.1.8 of Contract Part III(General Terms – Standard), the following requirement must be applied for escrow agreement and establishment. But, below could be subject to the discussion with customer and/or end-user.

♣ Fee for Escrow Service

E&M contractor is in charge of escrow agent fee including setup fee, annual fees. However, the cost of license and/or activities for the preparation of escrow package(including the license of software development tool) and activities for escrow verification testing shall be born by subcontractor.

If the update of Escrow package after deposit is needed due to software failure, the fee for updating shall be born by subcontractor.

Also, escrow agent can perform escrow verification testing. In that case, E&M Contractor is in charge of fees for escrow Verification Testing. However, the cost for preparation of this escrow verification testing shall be born by subcontractor.

♣ Selection of Escrow Agent

Supplier shall comply with end-customer's preferring escrow agent(e.g. Ironmountain) and comply with end-customer's presented escrow agreement document based on the selected escrow agent.

We can't accept supplier's preferring escrow agent and proprietary escrow agreement document in any case.

♣ Escrow Material List

The Escrow material to be deposited to Escrow Agent shall contain mandatorily as follows;

- Source code and related data file.
- Complier, link, design tool, proprietary tool etc. to generate application software from source code(must include of license information and transfer license to end user if certain terms and condition is reached) at supplier's expenses.
- Guideline document(build procedure) to generate application software from source code.
- Software Technical Document such as software design description and software requirement specification etc.
- Properly documented (architecture and planning, library software, operational system and settings,

communication protocols, files, etc)
- Others if requested by end-user

♣ Escrow Verification Testing

The Escrow Verification Testing Plan shall be submitted by subcontractor within 30 days before the testing.

For escrow verification testing, subcontractor has to have system or simulator for identifying software version number and performing function testing.

The Escrow Verification Testing Plan shall contain contents as follows;

The supplier shall identify:

A. where the verification is to take place
B. the personnel required
C. the expected duration of the verification exercise. Time must be included to verify the Escrow deliverables match the items identified in the Escrow Materials List
D. the procedures to be used to build the software (on of the items required in the Escrow Material List). These procedures will be followed to create the executables
E. the method to be used to verify that the information assembled results in the software executables are same to active software in trainset. This will depend on the software supplied and should be;
   - a comparison of checksums and program size and/or
   - a comparison of software version number after installation the compiled software into the system
   - execution of a defined set of tests to prove correct operation of the software
   - execution of a 'file compare' program or equivalent.

Otherwise, escrow agent can perform escrow verification testing as per customer's request in compliance with customer's escrow agreement document. Supplier must fully support escrow agent to perform escrow verification testing if requested without any cost implications.

4.5.4 Software Documentation

The following software documents based on IEEE-STD-1558(Type 5) shall be submitted.

- Software Project Management Plan
- Software Quality Assurance Plan
- Software Configuration Management Plan
- Software Verification and Validation Plan
- Software Verification and Validation Report
- Software Requirement Specification
- Interface Control Document
- Software Design Description
- Software Requirements Specification
- Software Requirement Traceability Matrix
- Software Test Plan
- Software Test Procedure
- Software Test Report
- Software Version Description Document

- Software User Manual
- Software Requirement Manual(CDRL #9-08)
- Master Software List
- Commercially Available Software List

4.6 Noise

Not Applicable

5. Design Information

In order to take an approval for the communication system from DENVER, the subcontractor shall provide all technical information as a written document. The below document/deliverable shall be provided as a minimum.

All technical information shall follow the "publications style guide" attached in TS.

5.1 General

The Subcontractor shall submit, the following general information in accordance with the schedule:

No	Document / Deliverables	Reference	*Deadline
1	Test and Inspection plan		At DDP
2	As-built drawing		At DDP
3	Operation & Maintenance Manual		At DDP
4	Illustrated parts catalog		At DDP
5	Training plan		At DDP
6	Training material		At DDP
7	List of warranty period spare parts		At DDP
8	List of recommended spare part		At DDP
9	List of special tools and test equipment		At DDP
10	PCB information		At DDP
11	Software Source code		At DDP

* Note: The deadline to be discussed and agreed between the Subcontractor and Hyundai-Rotem.

5.2 Design

The Subcontractor shall submit, the following design information in accordance with the schedule:

No	Document / Deliverables	Reference	*Deadline
1	Clause by Clause Compliance sheet for the Technical Specification of DENVER		Before Contraction

2	Dimensional Assembly Drawing by Autocad file.		At CDP
3	Sub-Assembly Drawing		At CDP
4	System description		At CDP
5	Function description		At CDP
6	Product description		At CDP
7	System Block Diagram or Schematic Diagram		At CDP
8	Installation Specification of On-board equipment		At CDP
9	Installation Specification and investigation report of Wayside equipment		At CDP
10	Interface Diagram and Specification with CDT		At CDP
11	Interface Diagram and Specification with Wayside Equipment		At CDP
12	Cabling requirement		At CDP
13	Test Specification		At DDP
14	Test Report		At DDP
15	Test Specification of Communication System at Complete Vehicle		At DDP
16	Test Report of Complete Vehicle		At DDP
17	Commissioning Test Procedure for Communication System (Including interface test with Wayside Equipment)		At DDP
18	Test Report of Commissioning Test		At DDP

* Note:

1. The deadline to be discussed and agreed between the Subcontractor and Hyundai-Rotem

5.3 SEM

The subcontractor shall provide all design information requested in the clause 4 of this PTS for EMI/EMC, Fire, RAMS and Software.

6. Tests and Inspection

The subcontractor shall perform all tests specified in the TS. The test procedure and the report shall be prepared and provided by the subcontractor.

7. Attachment

DENVER RTD EMU_ TS

CONTRACT PART1.pdf

Attachment 7 .pdf

DENVER RTD SEMS REQUIREMENT

Attachment_SEMS Requirement.zip